



Rafael Coimbra Azeiteiro

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About Me

Aerospace Engineering MSc student at Instituto Superior Técnico, Lisbon University, Portugal.

Master's dissertation research on Guidance, Navigation and Control (GNC) for atmospheric re-entry.

Technical Interests: Guidance, Navigation and Control (GNC), Satellite Formation Flying (SFF), Air-Breathing and Chemical Propulsion, Unmanned Aerial Systems (UAS) and Hypersonic Aerodynamics.

Education

Instituto Superior Técnico, Lisbon, Portugal - MSc in Aerospace Engineering

Sept 2024 - Today

- Major: Space ([Curriculum](#))
- Minor: Aerodynamics and Propulsion
- Awards: Academic Merit Board, 2024-2025

Instituto Superior Técnico, Lisbon, Portugal - BSc in Aerospace Engineering

Sept 2021 - Jul 2024

- Cumulative Grade: 16/20 ([Curriculum](#))

Projects

Master's Dissertation: GNC Architecture for Atmospheric Re-entry

- Developing rotating planet 3-DOF and 6-DOF simulation environments in MATLAB/Simulink for MSL-class Mars vehicles.
- Integrated NPCG guidance with logistic bank-angle parameterization to improve parachute deployment performance.
- Built a MATLAB linearization tool to extract local state space models to design a gain-scheduled LQR attitude controller.
- Implemented dead-reckoning state estimation using quaternion propagation, replicating realistic estimation conditions and sensor availability.

Member of [North Space](#) - 2026

- **BART Team Member, Autonomous UAV Systems (2026-Now)**– Developing a fully autonomous Pixhawk-based UAV platform, using MAVLink communication and offboard companion-computer guidance.
- **Astra Team Member, R&D Branch (2026-Now)** - Contributing to the development of a N₂O/ethanol liquid propulsion system for integration into a European Rocketry Challenge

(EUROC) rocket.

ISTrain (IMechE Railway Challenge)

- **Technical Director (2025)** - Leading a 30+ member railway student team in Instituto Superior Técnico. Responsible for strategy, system integration, sponsor relations and team engineering deliverables.
- **Vehicle Dynamics Team Lead (2025)** - Led a multidisciplinary subteam into the design and validation of the train's suspension system and pneumatic-actuated braking architecture.

Member of Aerotéc (Aerospace Engineering Students Group)

- **Website & Technologies Lead, Aerospace Week (2023)** - Organized Aerospace Week 2023 and managed the 2024 website ([Link](#)).
- **Rocket Workshop Member (2022-2023)** - Delivered workshops on rocket stability and design, guiding students through building and launching model rockets.

Experience

Summer Camp Counselor, Centro De Férias Desportivas Nos Mochos, Espinho, Portugal Jul 2024 - Sept 2024

- Summer Camp Counselor Volunteer at "Centro De Férias Desportivas Nos Mochos", dedicated to creating a fun, safe, and enriching environment for all campers aged from 6 to 15 years old.

Volunteer, Cruz Vermelha, Espinho, Portugal Since 2022

- Collected food donations from major chains to support poverty relief efforts.

Core Competencies

- C, Python (NumPy, SciPy, Orekit), Matlab/Simulink.
- MPC, SDRE/LQR/LQI, Linear Classical Control, Gain Scheduling, Monte Carlo campaigns.
- CAD (SOLIDWORKS), CFD (Star CCM+), Pneumatics (Festo FluidSIM), EM Analysis (CST Studio Suite, LTSpice).
- Requirements Engineering, Agile and Kanban Methodology.
- Git version contro, LaTeX, Excel, MS Office.

Soft Skills

 **Communication:** Public Speaking; Active Listening;

 **Collaboration:** Teamwork; Multidisciplinary Coordination; Conflict Resolution

 **Problem Solving:** Analytical Thinking; Root-Cause Analysis; Decision Making

 **Ways of Working:** Adaptability; Growth Mindset

Languages

- Portuguese - Native
- English - C1 (CAE, Grade B, Cambridge)